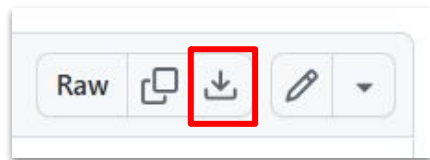


Lesson 6: Raster Data with rasterio

As you arrive and get set up...



1. Open **CryoCloud**: hub.cryointhecloud.com
 - Start a 1.9GB Python node
 - Create a folder named `lessons` (or any way you want to organize it)
2. Make a copy of **today's lesson notebook**:
 - In your web browser, visit:
https://github.com/NASA-SARP/2026-Coding-Alex/blob/main/lessons/06_raster_data.zip
 - Download the zip folder `06_raster_data.zip` and unzip:
 - “Drag and drop” the unzipped folder `06_raster_data` into the `lessons` folder you created earlier in CryoCloud
3. Also **download today's data file**:
https://github.com/NASA-SARP/2026-Coding-Alex/blob/main/lessons/06_raster_data/data/hls_20190531_4326.tif
 - We'll be uploading our data to CryoCloud as an exercise in today's lesson



Recap: From data types to data structures

Data types

Data type	Python name	What it holds	Examples
Integer	int	Whole numbers	14, 0, -3
Float	float	Decimal numbers	3.5, -0.2, 6.022e23
String	str	Text	"Jack", 'hello'
Boolean	bool	True or False	True, False

Data structures

List

```
mixed = ['abc', 34, True, 40, 'male']
```

DataFrame

	temperature	humidity	wind_speed
0	25	60	10
1	20	65	9
2	27	62	10
3	31	80	6

Dictionary

```
metadata = {  
    'location': 'Nairobi',  
    'quality_flag': 5,  
    'zenith': 60,  
    'clouds': True,  
}
```

GeoDataFrame

	COUNTYFP	NAME	geometry
0	001	Alameda	POLYGON ((-13612246.763 4538150.085, -13612346...
1	003	Alpine	POLYGON ((-13366503.508 4678946.015, -13366492...
2	005	Amador	POLYGON ((-13472697.863 4647652.232, -13472697...
3	007	Butte	POLYGON ((-13565004.793 4798394.124, -13564991...
4	009	Calaveras	POLYGON ((-13428575.483 4627725.227, -13428534...

Numpy ndarray

```
array([[0.0216, 0.0264, 0.0264, ..., 0.0228, 0.0228, 0.0218],  
       [0.0234, 0.0237, 0.0237, ..., 0.0228, 0.0228, 0.0218],  
       [0.0234, 0.0237, 0.0237, ..., 0.0187, 0.0187, 0.0199],  
       ...,  
       [0.482 , 0.482 , 0.4718, ..., 0.3391, 0.3268, 0.3358],  
       [0.4899, 0.4899, 0.4753, ..., 0.3253, 0.3248, 0.3809],  
       [0.4856, 0.4856, 0.4788, ..., 0.3048, 0.3667, 0.366 ]])
```

```
(7, 641, 847)
```

Numpy ndarray for raster data

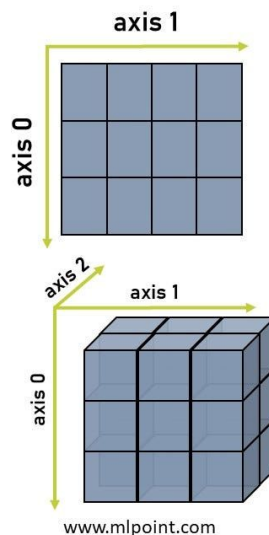
1	5	2	0
8	3	6	1
1	7	2	9

(3, 4)

(3, 4)

(3, 3, 2)

```
>>> a = np.array([[1, 2, 3, 4], [5, 6, 7, 8], [9, 10, 11, 12]])
>>> a
array([[ 1,  2,  3,  4],
       [ 5,  6,  7,  8],
       [ 9, 10, 11, 12]])
```



Stacking in NumPy Array

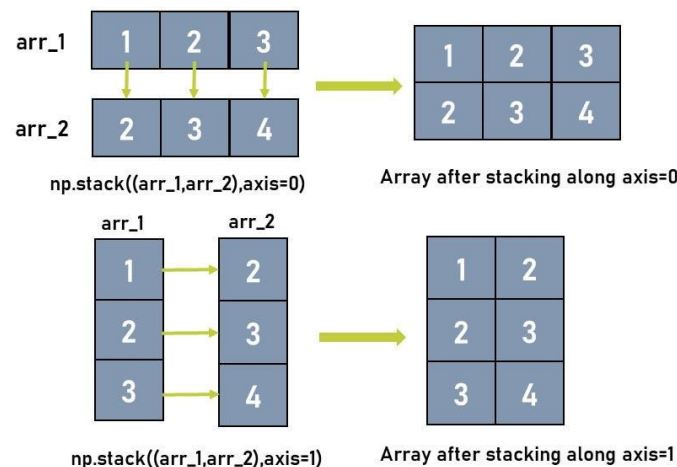
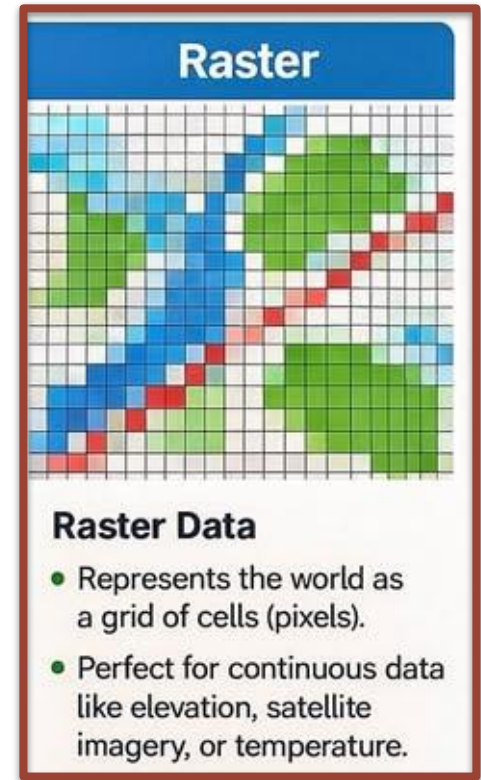
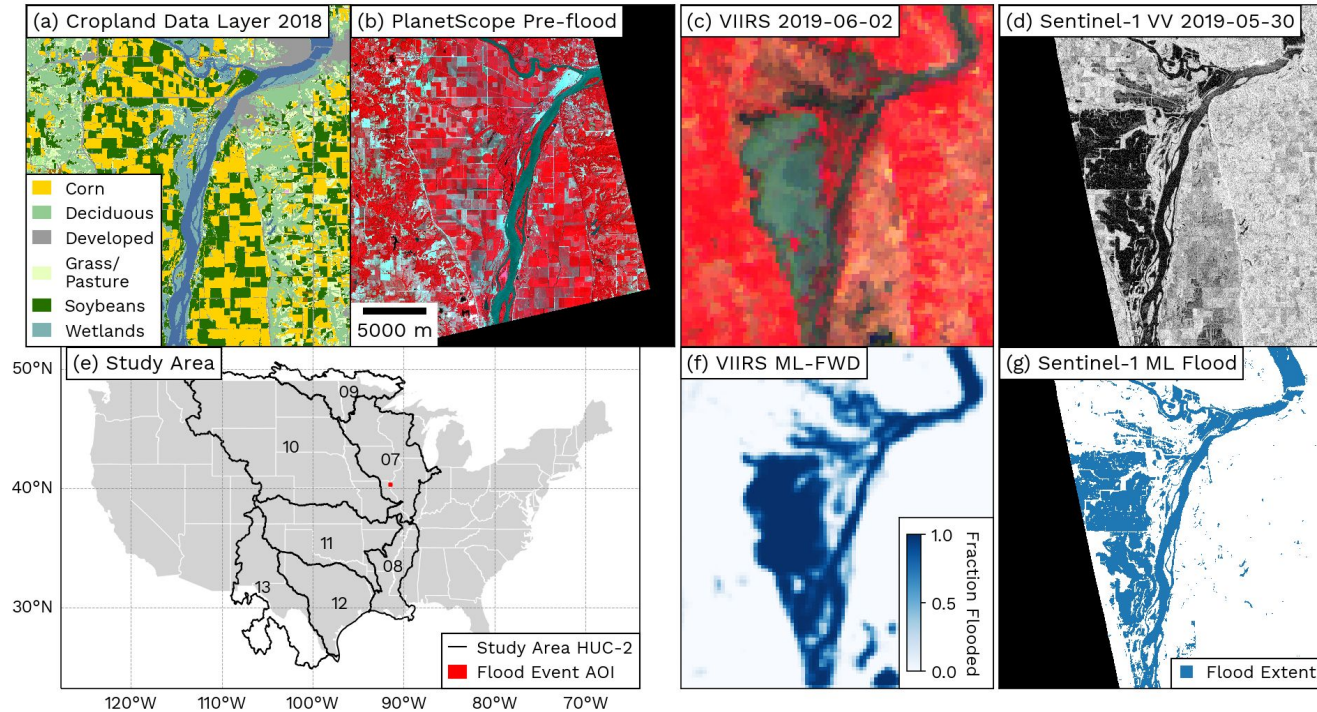


Figure: Paritosh Mahto Mipoint Medium

Types of spatial data



Harmonized Landsat Sentinel-2



Harmonized Landsat Sentinel-2



Raster metadata

```
{'driver': 'GTiff', 'dtype': 'float64', 'nodata': None, 'width': 847, 'height': 641, 'count': 7, 'crs': CRS.from_wkt('GEOGCS["WGS 84", DATUM["WGS_1984", SPHEROID["WGS 84", 6378137, 298.257223563, AUTHORITY["EPSG", "7030"]], AUTHORITY["EPSG", "6326"]], PRIMEM["Greenwich", 0, AUTHORITY["EPSG", "8901"]], UNIT["degree", 0.0174532925199433, AUTHORITY["EPSG", "9122"]], AXIS["Latitude", NORTH], AXIS["Longitude", EAST], AUTHORITY["EPSG", "4326"]'), 'transform': Affine(0.00026949458523585647, 0.0, -89.4808261250319, 0.0, -0.00026949458523585647, 37.1977986109348), 'blockxsize': 256, 'blockysize': 256, 'tiled': True, 'compress': 'lzw', 'interleave': 'pixel'}
```

Pixel width in units of the CRS

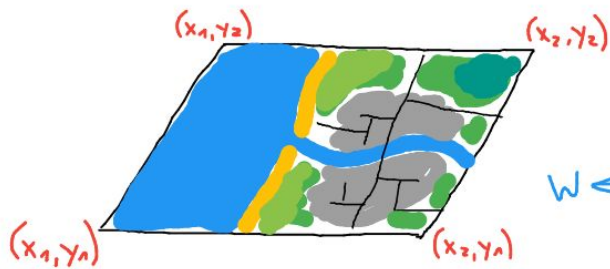
Upper left x

Affine(0.00026949458523585647, 0.0, -89.4808261250319, 0.0, -0.00026949458523585647, 37.1977986109348)

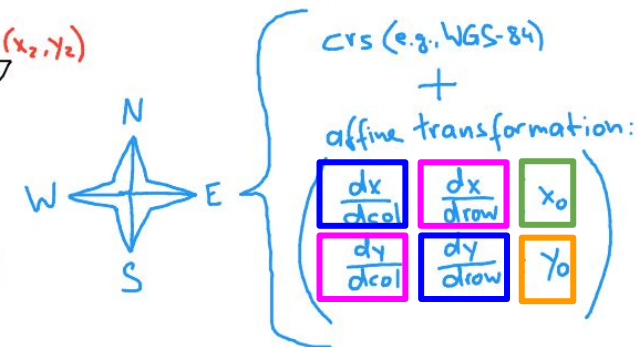
Rotation

Pixel height

Upper left y



bounds: $[x_1, x_2, y_1, y_2]$

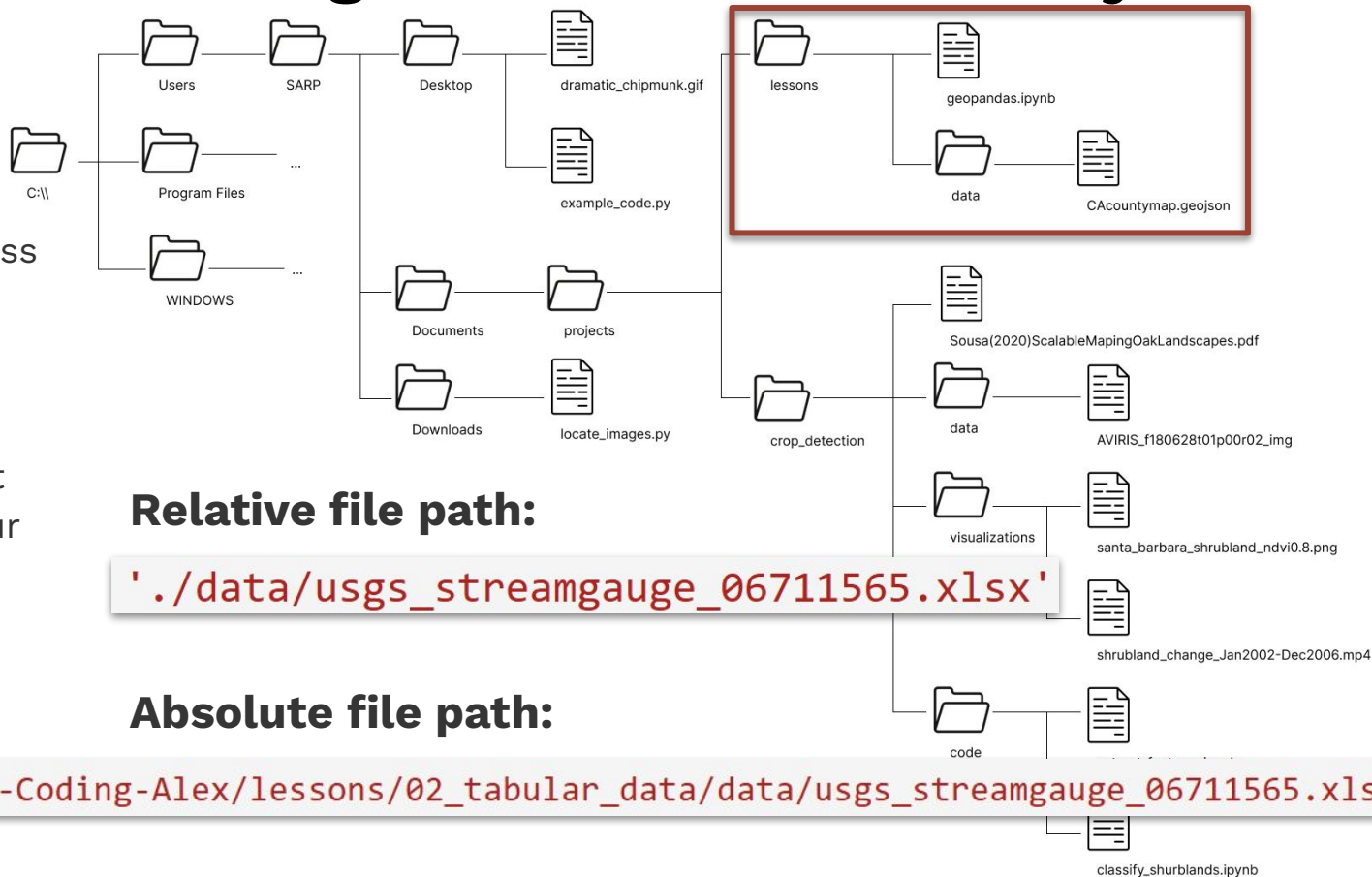


Michael Mommert via Medium

Affine allows us (rasterio) to:

- Convert pixels → coordinates
- Convert coordinates → pixels

Computers are organized “hierarchically”



Filepath: An address that specifies the location of any file on your computer

Root: The folder at the very top of your tree that contains everything

Relative file path:

`'./data/usgs_streamgauge_06711565.xlsx'`

Absolute file path:

`'/Projects/2026-Coding-Alex/lessons/02_tabular_data/data/usgs_streamgauge_06711565.xlsx'`

Lesson 6: Raster Data with rasterio

Lesson 6: Raster Data with rasterio

Please complete the **exit ticket**:

<https://forms.gle/UuvpwSEoqDbV5oGo9>

Don't forget to **shut down CryoCloud**:

File → Hub Control Panel → Stop Server